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Dark Patterns in Data-Consent Disclosures and Consumer Reactance to Online Behavioral Advertising

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ABSTRACT

This research investigates how dark patterns in data consent disclosures influence consumer responses to online behavioral advertising (OBA). Integrating the persuasion knowledge model (PKM), psychological reactance theory (PRT), and a cognitive accessibility perspective, we propose a framework explaining how manipulative consent designs shape perceptions of privacy threat, control, and persuasion outcomes. Across three studies, we demonstrate that ambiguous consent language and limited choice heighten perceived privacy threats and reduce perceived control, thereby activating persuasion knowledge and psychological reactance. Conversely, transparent designs enhance comprehension and autonomy perceptions. Targeted ads further act as retrieval cues that reactivate prior consent experiences, amplifying persuasion knowledge accessibility and resistance. These findings advance PKM and PRT by highlighting the cognitive accessibility pathway through which dark patterns influence consumer coping and persuasion resistance. The research offers theoretical and practical insights for designing transparent, autonomy-supportive consent interfaces that promote ethical and effective OBA.

Online behavioral advertising (OBA) is a dominant form of digital marketing that delivers personalized ads based on consumers' online behaviors and interests (Smit, Van Noort, and Voorveld 2014; Varnali 2021). Whereas this targeting often enhances engagement and conversion rates (Ham 2017), it depends heavily on the collection and use of personal data—typically acquired through website data consent interfaces. Regulatory frameworks like the California Consumer Privacy Act (CCPA 2018) and the General Data Protection Regulation (GDPR) of the European Union (EU) require explicit, informed consent before personal data can be collected or processed. However, in practice, the presentation of consent options often falls short of these standards.

A growing body of research highlights the ethical and legal concerns surrounding data privacy and consent, particularly in the digital advertising ecosystem (Boerman, Kruikemeier, and Zuiderveen Borgesius 2017, 2021, Boerman, Van Reijmersdal, and Rozendaal 2024; Ham 2017). Consumers frequently encounter consent requests without clear alternatives or transparent explanations, undermining the integrity of their choices. Increasingly, these practices are being recognized as examples of *dark patterns*—interface designs that intentionally mislead or coerce users into actions that benefit the platform at the expense of user autonomy (FTC 2022; Luguri and Strahilevitz 2021; Mathur, Kshirsagar, and Mayer 2021).

In the context of OBA, dark patterns often take the form of vague consent language (e.g., "I'm OK with that") or limited choice architecture (e.g., providing only an "Agree" button without an opt-out). These tactics exploit cognitive biases and reduce users' perceived control, prompting regulatory scrutiny and raising significant ethical questions. The Federal Trade Commission (FTC 2021, 2022) has explicitly warned against such practices, identifying them as a threat to informed consumer choice and privacy.

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Despite increasing regulatory attention, academic research in advertising has not fully addressed how dark patterns in consent disclosures affect consumer decision-making, psychological responses, and advertising effectiveness. This research seeks to fill that gap by examining how variations in consent interface design—specifically linguistic framing and choice architecture—affect perceived privacy threats, perceived control, and evaluations of targeted advertising. In doing so, we contribute to the growing discourse on ethical digital marketing and consumer well-being by providing an empirical foundation for understanding and mitigating the effects of dark patterns in OBA.

To address this research agenda, three studies were conducted, drawing on the persuasion knowledge model (PKM; Friestad and Wright 1994), psychological reactance theory (PRT; Brehm 1989), and the cognitive accessibility perspective (Keller 1987). Collectively, these studies examine how different dark pattern strategies in data consent disclosures influence consumer perceptions of subsequent OBA and how targeted ads, compared with random ads, moderate these effects. Study 1 explores consumers' understanding of consent language and recognition of manipulative intent, identifying "I am OK" as the most ambiguous and "I Agree" as the clearest label. Study 2 builds on these findings to test how consent language affects perceived privacy threats and control, revealing that including a decline option reduces perceived loss of control and psychological reactance. Study 3 extends this investigation by examining how targeted ads act as cues that heighten the accessibility of persuasion knowledge, drawing on cognitive accessibility research (Keller 1987; Souza and Oberauer 2016). Through this cueing process, targeted ads retrieve prior consent experiences from working memory, amplifying persuasion knowledge activation and reactance responses.

Our study makes a unique theoretical contribution by moving beyond the simple presence or absence of disclosures to examine how dark patterns in consent language affect the accessibility of persuasion knowledge and the activation of psychological reactance. Whereas prior research has treated disclosures as isolated events, we show that consumers' responses are shaped by their prior interactions with data consent disclosures. Specifically, we demonstrate that targeted ads—particularly those linked to cross-platform or third-party tracking—can serve as cues that reactivate memory of earlier consent experiences, heightening privacy vigilance. Our integrated framework offers a more dynamic account of how consumers recognize and resist manipulative data practices in digital advertising.

Literature Review

OBA and Data Privacy Regulations

Online behavioral advertising operates through advertising networks that track users' activities across multiple websites to deliver ads tailored to their interests (Ham 2017). By analyzing browser data, these networks monitor visits and page views to identify browsing sessions and store individualized information (Boerman, Kruikemeier, and Zuiderveen Borgesius 2017; Ham 2017). Despite its widespread use, users often lack a clear understanding of how OBA functions, its objectives, and the extent of data tracking (Boerman and Smit 2023; McDonald and Cranor 2010). McDonald and Cranor (2010) emphasized the importance of digital media literacy to help users make informed privacy decisions. Without adequate knowledge, consumers struggle to manage OBA effectively (Boerman, Kruikemeier, and Zuiderveen Borgesius 2017).

Although regulatory efforts have expanded, many publishers and advertisers still fall short of compliance with data-consent requirements, offering limited transparency or user control. Numerous privacy laws have been implemented or proposed worldwide, including the Digital Services Act and Digital Markets Act of the European Union (EU); the General Law for Data Protection in Brazil; the Digital Charter Implementation Act in Canada; Law No. 151 in Egypt; and the Protecting Kids on Social Media Act in the United States. Two landmark frameworks—the EU GDPR, implemented on May 25, 2018, and the U.S. CCPA, effective January 1, 2020—aim to safeguard personal data and regulate data use. The amended California Privacy Rights Act (CPRA, interchangeably *CCPA*), effective January 1, 2023, further mandates compliance for for-profit businesses that collect or process consumers' personal data, regardless of location, if they operate in California or gather data from California residents (California Privacy Protection Agency 2024). Under these laws, websites must clearly disclose how user data are collected and used and obtain informed, voluntary consent.

Data Consent Disclosures

Information disclosure plays a crucial role in digital marketing, prompting researchers to examine how such transparency helps consumers make better decisions. A significant milestone in data privacy is the FTC 2000 report to Congress, *Privacy Online: Fair Information Practices in the Electronic Marketplace*, which outlines key principles of fair information practices: *notice*, *choice*, *access*, and *security*. Scholars of advertising have focused on the principle of *notice*, particularly analyzing the FTC Clear and Conspicuous Standards (Hoy and Andrews 2004; Hoy and Lwin 2007). Additionally, the FTC (2013) guidelines in *.com Disclosures* emphasize that information must be presented in a way that is clear, prominent, and easily accessible to guide informed decisions. The principle of *choice* requires that users be given control over how their data are used. Similarly, the CCPA mandates that marketers clearly define their objectives, such as targeted advertising, and explain their data collection methods, including the use of cookies. These data practices must allow consumers to make informed and voluntary choices, with consent being *freely given, specific, informed*, and *unambiguous* (CCPA 2018).

Utz et al. (2019) made a notable contribution to understanding consent design through research conducted just before the CCPA took effect. Focusing on the GDPR in Germany, they examined how notice placement, choice structure, and content framing within graphical interfaces influence user consent. The study analyzed consent formats—*no option*, *confirmation only*, *binary* (*accept* or *decline*), and *checkbox for essential versus marketing purposes*—and found that 57.4% of the notices contained nudging or dark pattern elements that promoted privacy-unfriendly choices. These elements included highlighting default privacy-invasive options, concealing advanced settings in obscure links, and pre-selecting data collection boxes. Based on these findings, Utz et al. (2019) emphasized the need for stricter regulatory guidance to ensure that consent interfaces support informed and autonomous user decisions.

Recent literature on data privacy regulations underscores the importance of transparent consent disclosures in fostering trust around OBA. Yet, multiple studies have shown that such disclosures often employ misleading or manipulative tactics (Miller 2021; Padejski 2021). The FTC (2023) reported that only 20% of reviewed websites complied with fair information practices, raising substantial privacy concerns. Although the CCPA requires marketers to inform consumers about how their data are used for targeted advertising, little is known about how users interpret and respond to manipulative consent language. Addressing this gap, Study 1 serves as an exploratory step to examine how consumers perceive commonly used consent labels and identify which phrases are viewed as ambiguous or manipulative—key features of dark patterns. This investigation provides a foundation for understanding the psychological effects of consent language and for shaping the experimental designs that follow. Accordingly, Study 1 addresses the following research question:

> **Research Question 1:** How do consumers perceive the dark patterns in data consent disclosures?

Theoretical Framework

PKM and PRT

Since its introduction, the PKM (Friestad and Wright 1994) has evolved to encompass broader applications, leading to the development of conceptual persuasion knowledge (CPK; Boerman, Van Reijmersdal, and Neijens 2012; Boerman et al. 2018; Boerman, Van Reijmersdal, and Rozendaal 2024). *Conceptual persuasion knowledge* refers to consumers' ability to recognize persuasive intent, tactics, and potential consequences by identifying the underlying motive of a persuasion attempt. Activation of CPK occurs when consumers recognize these persuasive tactics (Boerman et al. 2018; Boerman, Van Reijmersdal, and Rozendaal 2024). In this study, dark patterns are conceptualized as covert persuasion tactics designed to obtain user consent for data collection, thereby activating CPK. When consumers identify these manipulative designs as intentional nudges, CPK activation promotes resistance behaviors such as declining or withholding consent. Conversely, when dark patterns are not recognized as persuasive tactics, CPK remains inactive, and consumers are more likely to grant data consent. This proposition aligns with the core premise of PKM that persuasion knowledge is triggered by the recognition of persuasive intent in hidden or manipulative message designs.

We suggest that dark patterns may provoke reactance when individuals recognize persuasive tactics at play. According to PRT, when people perceive their autonomy is being constrained, they experience psychological arousal, which triggers a motivation to reclaim that freedom (Brehm 1966; 1972; Brehm and Brehm 1981). This concept has been widely applied in persuasion studies, including coupon use (Trump 2016), exclusive loyalty programs (Kivetz 2005; Wendlandt and Schrader 2007), and personalized advertising (Baek and Morimoto 2012; Brinson, Eastin, and Cicchirillo 2018). Providing consumers with transparent and balanced choices about sharing their data with advertisers can enhance their sense of control over privacy. Offering clear explanations of data collection practices, along with both approval and rejection options, helps reassure users that they retain autonomy over their personal information. This approach reduces perceived threats to freedom of choice. In contrast, the use of dark patterns—such as vague or manipulative consent mechanisms—can make consumers feel their freedom is undermined. As a result, they may resist the intended actions or respond negatively to data-driven advertising efforts.

To be specific, using clear and simple language in data consent disclosures can help individuals better grasp the persuasive intentions and underlying motives during the consent process. This method may enhance their cognitive access to existing persuasion knowledge, making it easier for them to identify such intentions (Campbell and Kirmani 2000; Ham and Nelson 2019; Isaac and Grayson 2017; Wright, Friestad, and Boush 2005). Previous research indicates that recognizing persuasive intent involves making consumers aware of the message's persuasive nature, such as in native advertising or sponsored content, through straightforward labels, such as "advertisement" (Ju, Lee, and Sherrick 2022). However, the existing research does not provide sufficient understanding of how consumers perceive dark patterns as manipulative tactics that infringe upon their autonomy.

Persuasion Knowledge and Reactance Activation Processes

When individuals perceive a threat to their autonomy, they often engage in behaviors aimed at restoring their freedom (Brehm 1966; Quick and Stephenson 2008). In personalized advertising, this autonomy is closely tied to consumers' desire to protect their privacy from entities that track online behavior for targeted promotions (Brinson, Eastin, and Cicchirillo 2018). According to Brehm's (1993) theory of control, perceived privacy control reflects the extent to which individuals feel they can voluntarily share personal data without external pressure. Transparent, clearly worded consent mechanisms help maintain this sense of control, whereas vague or manipulative disclosures can elicit *psychological reactance*—a motivational state aimed at restoring perceived freedom (Brehm 1966, 1972).

Vague consent language such as “I am OK” can act as a subtle form of coercion by obscuring the implications of data sharing and offering no meaningful choice, especially when opt-out options are absent. Such wording limits agency, nudges users toward consent, and conceals persuasive intent, thereby threatening autonomy. In contrast, clear phrasing such as “I Agree” communicates transparency and allows deliberate, informed choice—reducing perceptions of manipulation and enhancing perceived control (Boerman, Kruikemeier, and Zuiderveen Borgesius 2017; Boerman et al. 2018, 2021; Boerman, Van Reijmersdal, and Rozendaal 2024; Ham 2017).

Integrating the PKM and PRT provides a framework for understanding how consumers interpret and respond to consent environments. When dark patterns are recognized as persuasive tactics, CPK becomes activated, making the manipulative intent salient. This awareness increases perceptions of coercion and subsequently triggers psychological reactance, as consumers attempt to restore autonomy and resist influence. Reactance, in turn, reduces message persuasiveness by undermining attitudes and behavioral intentions (Dillard and Shen 2005; Quick et al. 2015; Reinhart et al. 2007; Zhang 2020). Prior research shows that high reactance often leads to message rejection or source avoidance (Bleier and Eisenbeiss 2015; Rains 2013). Moreover, as Schwartz and Solove (2011) note, control over personal data is widely regarded as a fundamental right; thus, vague or coercive consent practices intensify perceived threats to this right. Providing explicit, meaningful choices through clear language and opt-out options can restore perceived autonomy and mitigate reactance (Brehm 1966, 1972; Brinson, Eastin, and Cicchirillo 2018; Dillard and Shen 2005).

In the context of OBA, obtaining user consent is the first step in delivering targeted advertising. Consumers' recognition of dark patterns in consent disclosures is expected to activate CPK, which heightens reactance and influences subsequent attitudes toward the website, brand, and purchase intention (PI) (Boerman et al. 2018; Boerman, Kruikemeier, and Zuiderveen Borgesius 2021; Boerman, Van Reijmersdal, and Rozendaal 2024).

In summary, vague consent labels such as “I am OK” are likely perceived as deceptive and coercive, heightening privacy concerns and reducing perceived control. In contrast, clear consent language such as “I Agree” signals transparency, reinforces autonomy, and fosters more favorable responses toward the website, brand, and PI. Based on this theoretical integration and prior evidence, we propose the following hypothesis:

> **Hypothesis 1:** Compared with vague consent language (“I am OK”), clear consent language (“I Agree”) will lead to more positive attitudes toward (a) the website, (b) the brand, and (c) higher purchase intention, by reducing perceived threats to data privacy and increasing perceived control over personal information.

Targeted OBA as a Moderator: Cognitive Accessibility Cue

Research on the PKM emphasizes that the accessibility of persuasion knowledge is essential for its activation (Boerman, Van Reijmersdal, and Neijens 2012). *Accessibility* refers to how readily consumers can retrieve knowledge about persuasive intent, often assessed by whether they recognize an ad's persuasive purpose (Wojdynski and Evans 2016) or a salesperson's tactics (Campbell and Kirmani 2000). Recognizing persuasion requires both noticing a message and understanding its underlying intent, highlighting the importance of accessible cognitive frameworks in shaping awareness. This notion parallels memory research in advertising (Keller 1987), which suggests that stored information influences later judgments and behaviors—even when substantial time has passed between exposure and decision.

Whereas most prior work has examined how advertising messages are encoded and retrieved from memory, the present research extends this inquiry to how consumers may recall prior data consent experiences when later exposed to related advertising. In this context, targeted ads may serve as retrieval cues that reactivate memory traces associated with earlier encounters, including the consent process. Because targeted ads are connected to a user's previous interaction with a specific website, they may prompt consumers to mentally revisit that earlier decision. For instance, when an ad from the same website appears after a user has granted data permission, it may remind the user of that prior disclosure, drawing attention to how and why personal data were collected. In this way, targeted ads may operate not only as persuasive messages but also as *retrospective cognitive cues* that increase awareness of persuasive intent.

This perspective builds on associative network models of memory, which conceptualize knowledge as interconnected nodes that spread activation among related concepts (Anderson 1983; Keller 1987). Retrieval depends on the strength and structure of these associations: recall is more likely when the context at retrieval resembles the context at encoding (Alba and Hutchinson 1987; Bettman 1979; Lynch and Srull 1982). When an ad references the same brand or service linked to an earlier consent, such contextual overlap may strengthen associative links and enhance memory accessibility (Boerman, Van Reijmersdal, and Rozendaal 2024; Ham 2017).

Integrating these insights with PKM and PRT, we propose that targeted OBA may act as a cognitive retrieval cue that enhances the accessibility of persuasion knowledge. Within the PKM framework, persuasion knowledge activation depends not only on availability but also on accessibility (Boerman, Van Reijmersdal, and Neijens 2012; Campbell and Kirmani 2000). However, accessibility can be constrained when dark patterns are subtle or when consumers fail to recall earlier consent interactions. Because targeted ads often mirror the original context in which data were shared, they may facilitate retrieval of that prior experience, making manipulative intent more salient and thereby activating CPK.

This process aligns with the *retro-cue effect* in cognitive psychology, which shows that cues presented after an encoding event can direct attention within working memory and improve recall (Souza and Oberauer 2016). In the current context, a targeted ad may operate as such a retro-cue, heightening awareness of the persuasive tactics embedded in earlier data consent interactions. In contrast, non-targeted ads—unrelated to prior browsing—are less likely to trigger these associations, making the recall of prior consent and activation of persuasion knowledge less probable.

Taken together, this integration extends PKM and PRT by introducing *cognitive accessibility* as a linking mechanism that connects prior consent experiences with responses to subsequent advertising. Specifically, we propose a moderated mediation process in which dark pattern consent tactics (e.g., the absence of a decline option) influence perceived privacy threats and perceived control, which in turn affect ad evaluations. These effects are expected to be stronger when targeted ads make prior consent experiences more cognitively accessible and thereby activate persuasion knowledge and psychological reactance. When a targeted ad from the same source follows a manipulative consent interface, perceptions of intrusion and privacy violation may become more salient, amplifying concerns about autonomy and control. On this basis, we advance the following hypothesis:

> **Hypothesis 2** (*moderated mediation*): The effect of consent option type (agree-only vs. agree + decline) on (a) website attitudes, (b) ad attitudes, (c) brand attitudes, and (d) purchase intention will be sequentially mediated by (1) greater recognition of persuasive cues, (2) heightened perceived privacy threat, and (3) reduced perceived control. This indirect pathway will be moderated by ad type, such that exposure to targeted ads (vs. non-targeted ads) enhances recognition of the prior consent disclosure, thereby strengthening the sequential indirect effect on attitudinal and behavioral outcomes.

Overview of Studies

The three studies were approved by the Purdue University Institutional Review Board (IRB-2024-782), and all participants provided informed consent. We began with a qualitative inquiry (Study 1) aimed at understanding how consumers perceive and react to various types of data consent language (e.g., "Agree," "I am OK"). These insights laid the foundation for our integrated framework that explains how certain message features in data consent disclosures trigger persuasion knowledge and psychological reactance. Building on these findings, Study 2 experimentally tested how the wording of consent language influences consumers' perceptions of data privacy threats and their sense of control, both key components of the reactance mechanism. In both Studies 1 and 2, participants were exposed only to simulated websites, and thus the dependent measures focused on attitudes toward the website, brand, and PI. In Study 3, we expanded the model by introducing advertising stimuli to assess whether exposure to targeted ads could increase retrospective recognition of the data consent disclosure and moderate subsequent perceptions and evaluations. This approach allowed for the inclusion of attitude toward the ad (Att_{ad}) as an outcome variable, supporting a progressive, theory-driven research design. Figure 1 presents the expanded conceptual framework.

Study 1

Method

Participants

We employed CloudResearch to recruit adult Internet users residing in the United States. Prior studies have shown that using a verified sample from CloudResearch yields high-quality data (Berry et al. 2022). Our exploratory study (Study 1) included 122 U.S. residents. Participant age ranged from 22 to 76 years, with 59.5% male and 40.5% female. In terms of race/ethnicity, 74.4% identified as White, followed by 15.7% as Black or African American, and 6.6% as Asian. Regarding income, 24.8% reported \$35,000 to \$54,999, 23.1% reported \$55,000 to \$74,999, and 12.4% reported \$75,000 to \$94,999. Furthermore, 55.4% of participants had a college degree.

Procedure and Instruments

To explore Research Question 1, we investigated consumers' comprehension and assessment of data consent disclosure language. We reviewed prominent U.S. company websites (e.g., Fortune 500 companies) to glean initial insights into common data consent labels, identifying eight prevalent labels. To gauge consumer perceptions of each label, respondents were randomly assigned to evaluate a specific label on a fictitious travel company website, after which we measured their cognitive responses by prompting them to express their thoughts and feelings regarding the data

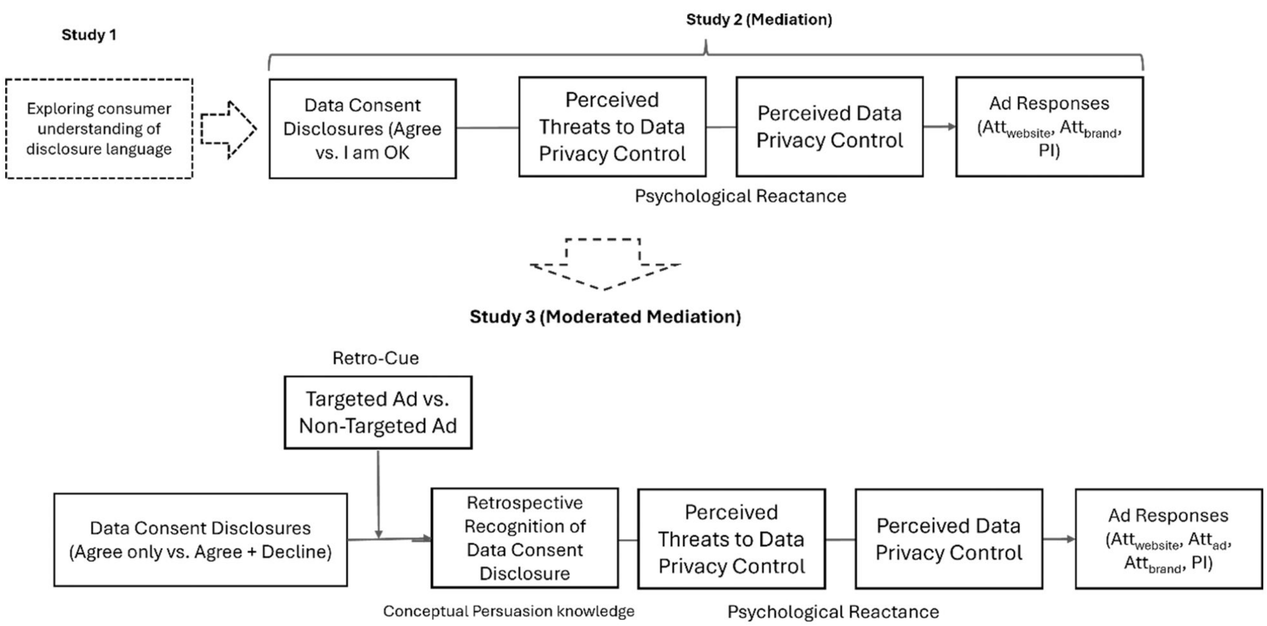

Figure 1. Stepwise progression of the conceptual framework across studies.
Note. Att_{ad} = attitude toward the ad; Att_{brand} = attitude toward the brand; $Att_{website}$ = attitude toward the ad; PI = purchase intention.

consent disclosure. Cognitive responses to an open-ended inquiry regarding data consent disclosure were assessed and examined through both qualitative and quantitative analysis techniques.

We manipulated eight distinct data consent disclosures, maintaining uniformity in visual elements and word counts across all versions, with the exception of the data consent labels. Variations in the labels included "accept," "agree," "I like Cookies," "I Am OK," "I Am Happy," "Continue," "Got It," and "Dismiss." After the presentation of stimuli, participants were instructed to share their cognitive thoughts and feelings. Respondents answered the question: "Please indicate in as much detail as possible what characteristics of the data consent notice led you to believe that it was deceptive or non-deceptive." A total of 122 responses were collected, focusing on the respondents' recognition and evaluations of consent language utilized in the data consent disclosures (see Appendix A).

Data Analysis and Preliminary Findings

We examined how consumers interpret and respond to data consent language using our integrated framework. The analysis revealed that perceived ambiguity was the primary factor influencing whether individuals viewed a label as indicative of dark patterns. The absence of a decline option also played a critical role. From a regulatory perspective, these practices align with definitions of dark patterns outlined by the CCPA (2018) and the FTC (2013, 2022). Overall, participants demonstrated notable recognition of and resistance to manipulative consent features. Several participants explicitly identified the lack of opt-out options as coercive, describing how being presented only with a notification or acceptance choice made them feel pressured to comply.

Participants frequently expressed that unclear or conversational phrasing made consent language deceptive. One participant noted:

> The way it was worded, "I like cookies" could cause people to not really understand what it means. It was tricky by using the words "I like" in the consent instead of just "accept or decline."

Another respondent emphasized the need for genuine choice and transparency:

> Users should be given a real choice to accept or reject cookies. A non-deceptive notice would not only have an "Accept" button but also a "Reject" or "Manage settings" option. If only an "Accept" option is provided, it could be considered deceptive. Non-deceptive notices should respect the user's choice. If it's hidden, uses complex language, or is difficult to understand, it could be seen as deceptive.

These comments underscore a critical point: data consent labels must meet clarity standards mandated by the FTC (2022). When ambiguous language diverged from explicit terms (e.g., I Agree), participants perceived such phrasing—particularly when paired with missing opt-out options—as potentially deceptive or manipulative. Some participants also expressed uncertainty about whether the language was deceptive, highlighting confusion as an element of perceived coercion:

> I honestly could not tell if it was deceptive or not!

This sense of uncertainty and coercion aligns with FTC and CPRA guidelines, which define dark patterns as practices that obscure users' ability to make informed, voluntary choices. Collectively, participants' interpretations reflected concerns about misleading design, unclear language, and limited control over personal data. They generally perceived clear consent language (e.g., I Agree) as less deceptive and ambiguous phrasing (e.g., I am OK with that) as more manipulative. Overall, while participants recognized some deceptive elements, their vigilance toward consent practices appeared limited. These insights informed the formal coding procedure implemented in the next phase.

Perceptions of Dark Patterns

Expanding upon initial findings, we compared qualitative insights with existing literature on dark patterns. Because some participants did not adequately distinguish consent labels, we conducted quantitative content analysis on their cognitive responses. In this study, perceptions of misleading or deceptive consent practices—such as vague language or the absence of opt-out options—are conceptualized as perceived instances of dark patterns. This consistent definition reflects both regulatory and academic perspectives and is applied throughout the manuscript to maintain conceptual precision. The phrases may include *coercive*, *deceptive*, *malicious*, *misleading*, *obnoxious*, *seductive*, *steering*, *trickery*, *confuse*, *deceive*, *exploit*, *manipulate*, *mislead*, *subvert*, *undermine*, *without user consent*, *without user knowledge*, *harm*, *obfuscate*, *distract*, and *abuse* (Mathur, Kshirsagar, and Mayer 2021). We conceptualized responses containing phrases indicative of dark patterns as such.

We also identified unlisted exceptions, where participants expressed recognition of dark patterns using phrases not covered in the inventory. For example, participants found the absence of opt-out options and insufficient explanations about cookies unfavorable. Additional words suggesting a sentiment associated with dark patterns (e.g., "suspicious," "tried to get me to click") were identified. These cases were categorized as dark patterns, indicating the participant's recognition of improper data consent practices.

Two trained coders independently reviewed all open-ended responses and applied a binary coding scheme: responses indicating the presence of dark patterns were coded as 1, and those not referencing dark patterns were coded as 0. Prior to formal coding, coders were trained using established definitions of dark patterns drawn from the literature and conducted a pilot session to align on coding criteria. Discrepancies were resolved through discussion, resulting in high intercoder reliability (Perreault and Leigh's Ir = .98) (Perreault and Leigh 1989). Of the total responses ($N = 122$), approximately 40.2% ($n = 49$) were coded as referencing dark patterns. When applied to the specific consent label pairs, results revealed that participants clearly distinguished between "I Agree" and "I am OK with that."

After providing their open-ended responses, participants were briefly introduced to the concept of dark patterns in data consent practices. They then rated their agreement with the following statement assessing their understanding and perception of such practices using a 7-point Likert scale (1 = *strongly disagree*, 7 = *strongly agree*), for which higher scores indicate stronger perceptions of dark patterns and lower scores indicate weaker perceptions:

> Dark patterns in digital marketing are frequently portrayed as offering consumers choices regarding privacy settings or data sharing, but they are deliberately crafted to steer consumers toward disclosing more personal information?

This follow-up question enabled a quantitative assessment of the participant's endorsement of the dark pattern definition and validated the consistency between their open-ended reflections and structured recognition of manipulative consent design. Specifically, "I Agree" was more frequently perceived as an explicit and transparent label, whereas "I am OK with that" elicited higher perceptions of implicit manipulation ($F(1, 119) = 7.13$, $p = .009$, $\eta_p^2 = .06$). These findings complement the qualitative insights and address Research Question 1 by illustrating how consumers interpret and respond to different forms of data consent language.

Study 1 Discussion

Study 1 showed that participants frequently perceived ambiguous or vaguely worded consent labels as dark patterns, indicating that such interface elements can activate consumers' CPK and prompt resistance to perceived persuasion attempts. This finding underscores the importance of perceived transparency and intent in shaping evaluations of digital data practices. Consistent with PKM and PRT, clearer and more explicit consent labels appear to reduce perceived manipulativeness and promote more informed user decisions.

Beyond revealing consumer concerns about label clarity, Study 1 provided essential theoretical and methodological groundwork for the broader framework. Whereas prior PKM research has focused on consumers' stored knowledge of persuasion tactics, relatively little attention has been given to how contextual cues—such as consent interfaces—trigger recognition of persuasive intent. Our findings suggest that these cues are meaningful stimuli in everyday privacy management and play a unique role in activating persuasion knowledge.

Accordingly, Study 1 served as an exploratory foundation for the experimental design that followed. It enabled the empirical identification of naturally occurring consent labels differing in perceived manipulativeness, rather than relying on researcher-imposed classifications. This consumer-centered approach aligns with the emphasis of PKM on persuasion from the audience's perspective. Building on these insights, Study 2 tests how two consent labels—"I Agree" and "I am OK with that", representing *minimal* and *maximal* dark pattern perceptions, respectively—influence perceived privacy threat and perceived control as part of a broader psychological reactance process.

Study 2

Method

Participants

Participants were adult Internet users residing in the United States, recruited through CloudResearch. We recruited 200 adult participants to ensure sufficient sample size and allow for potential exclusions during data cleaning. Participants ranged in age from 18 to 80 years. The sample was 50.8% male and 48.7% female. Most identified as White (78.5%), followed by Black or African American (9.7%) and Asian (7.2%). Income levels were diverse: 22.6% reported annual earnings of $35,000–$54,999, 15.4% earned $100,000–$149,999, and 14.9% earned $55,000–$74,999. Most participants (58.5%) held a college degree.

Experimental Design, Stimuli, and Procedure

Building on Study 1, we conducted a between-subjects experiment manipulating consent language ("I Agree" vs. "I am OK with that"). After providing electronic informed consent, participants completed the Qualtrics survey. As in Study 1, they viewed a fictional travel agency website—selected because travel represents a common and relatable consumer context and has been widely used in prior OBA research (e.g., Lee and Rha 2013; Segijn and van Ooijen 2022). Participants were randomly assigned to one of the two consent language conditions. After reviewing the website, they reported their attitudes toward the website and brand, PI, perceived privacy threat, and perceived privacy control. Finally, demographic information was collected, and participants were debriefed and thanked.

Measures

Attitudes toward the website ($M = 4.05$, SD = 1.38, ω = .95), brand ($M = 4.15$, SD = 1.39, ω = .97), and PI ($M = 3.22$, SD = 1.68, ω = .97) were measured using five bipolar evaluative items on a 7-point scale (modified from MacKenzie, Lutz, and Belch 1986), which assessed attitudes (*unappealing/appealing; good/bad(r); unpleasant/pleasant; favorable/unfavorable(r); unlikeable/likeable*) and PI (*very unlikely/very likely*; *very unwilling/very willing*; *very uninclined/very inclined*). To measure perceived threats to data privacy, we used the following four items ($M = 4.42$, SD = 1.58, ω = .93) on a 7-point Likert type scale (modified from Dillard and Shen 2005; Miller et al. 2007): "*The cookie consent notification threatened my freedom to choose.*"; "*The data consent notification tried to make a decision for me.*"; "*The data consent notification tried to manipulate me.*"; and "*The data consent notification tried to pressure me*". Perceived privacy control ($M = 3.72$, SD = 1.47, ω = .91) was measured using a 7-point Likert type scale for the following four items (Esmark, Noble, and Breazeale 2017): "*I was completely satisfied in my ability to control the level of privacy I had on the website..*"; "*I had little reason to be concerned about my privacy on the website..*"; "*I controlled my level of privacy on the website.*"; and "*I had control over who I dealt with.*").

Results

Manipulation and Randomization Checks

To avoid any potential bias that could influence responses on the dependent variables, we conducted the manipulation check in a separate pretest independent of the main study. In this pretest, participants viewed the same travel websites used in the main experiment, which differed only in the data consent labels presented ("I am OK with that" vs. "I Agree"; see Appendix A). After being briefly introduced to the concept of dark patterns, participants rated the extent to which the website's data consent interface reflected such practices using a 7-point Likert scale. Results confirmed that participants perceived "I am OK with that" as embodying more characteristics of dark patterns ($M = 5.73$, SD = 1.22) than "I Agree" ($M = 4.37$, SD = 1.71; $F(1, 29) = 6.40$, $p = .02$, $\eta_p^2 = .18$).

In the main study, participants were randomly assigned to the same two conditions without any mention of dark patterns, thus ensuring that their evaluations of the dependent variables were not influenced by prior conceptual priming. Group assignment was balanced (I am OK: 49.5%, I Agree: 50.5%), and randomization checks confirmed no significant differences across experimental groups on key demographics, including age ($F(1, 198) = 2.00$, $p > .05$); gender ($\chi^2(1, N = 200) = .001$, $p > .05$); education ($F(1, 198) = .00$, $p > .05$); income ($F(1, 198) = .09$, $p > .05$); and race ($\chi^2(5, N = 200) = 3.48$, $p > .05$).

Hypothesis Tests

We employed SPSS PROCESS to test Hypothesis 1_a ($Att_{website}$; $M_{i\text{-}agree} = 4.12$ vs. $M_{i\text{-}am\text{-}ok} = 3.94$), Hypothesis 1_b (Att_{brand}; $M_{i\text{-}agree} = 4.00$ vs. $M_{i\text{-}am\text{-}ok} = 3.86$), and Hypothesis 1_c (PI; $M_{i\text{-}agree} = 3.36$ vs. $M_{i\text{-}am\text{-}ok} = 3.55$), which together examined whether perceived threats to data privacy and perceived data privacy control function as serial mediators in the relationship between data consent disclosure language and ad-related evaluations. Although the descriptive means did not show a strictly linear pattern across conditions, such variability is characteristic of complex mediation models, for which effects primarily operate through indirect rather than direct relationships. We used 5,000 bootstrapping resampling and 95% confidence interval (CI) (Model 6). *Heteroscedasticity-consistent*inference was made using HC4 (Cribari-Neto) adjustments (Hayes and Cai 2007). The results showed that the effect of data consent disclosure language on $Att_{website}$ was serially mediated through the mediation path ($b = .10$, Boot*SE* = .05, 95% CI [.0186, .2063]), excluding zero. The direct effect of language on $Att_{website}$ was not significant ($b = .14$, *SE*(HC4) = .16, 95% CI [−0.1807, .4692]), including zero. The results showed *indirect-only mediation* suggesting that the mediator identified was consistent with the hypothesized theoretical framework (Zhao, Lynch, and Chen 2010). In the same vein, Att_{brand} was also serially mediated through the same path (b = .11, SE = .05, 95% CI [.0186, .2240]), excluding zero. The direct effect of language on Att_{brand} was not significant (b = .17, SE(HC4) = .19, 95% CI [–0.2086, .5554]), including zero. The results showed *indirect-only mediation*. The same pattern was found for PI (b = .13, SE = .06, 95% CI [.0230, .2643]), excluding zero. The direct effect of language on PI was not significant (b = −0.12, SE(HC4) = .19, 95% CI [−0.4965, .2673]), including zero. Overall, the data reveal that when dark patterns—such as vague consent labels—are present in data consent disclosures, they increase perceived threats to data privacy, which in turn diminish perceived control. As a result, participants responded more negatively across all evaluation outcomes, confirming support for Hypothesis $1_{a–c}$. The indirect and direct effects are presented in Table 1.

Study 2 Discussion

Study 2 demonstrated that the linguistic framing of data consent language shapes consumers' perceptions of persuasive intent. Vague phrasing such as "*I am OK with that*" heightened perceived privacy threats and reduced perceived control, consistent with our integrated framework. Across Studies 1 and 2, participants also noted that manipulative consent practices involve not only language but also choice architecture—the structure of the options presented. The absence of an opt-out button (e.g., "*Decline*") was widely interpreted as a dark pattern, activating CPK and signaling a lack of transparency and fairness.

Building on these insights, Study 3 shifts focus from linguistic framing to structural manipulation, examining how limited consent options—interfaces that present only an "I Agree" button—constrain user autonomy and reflect a dark pattern in digital marketing. By comparing "I Agree Only" and "I Agree and Decline" designs, Study 3 extends this inquiry to more realistic, ecologically valid consent structures.

We further propose that restrictive choice architecture may heighten awareness of persuasive intent, particularly when reinforced by targeted ads that act as cognitive retrieval cues, making prior consent experiences more salient. In this way, Study 3 integrates interface design and cognitive accessibility perspectives to explain how structural dark patterns activate persuasion knowledge and resistance to behavioral advertising.

Together, these studies advance the understanding of dark patterns as multidimensional—spanning both linguistic framing and structural design—and align with evolving FTC guidance (2013, 2022), which recognizes manipulative practices that obscure or limit user choice. Study 3 therefore addresses a critical gap by empirically testing how common consent architectures shape consumer autonomy and perception under this broader regulatory framework.

Study 3

Method

Participants

We employed CloudResearch to recruit adult Internet users residing in the United States. To allow for data cleaning and potential invalid responses, 400 participants were initially recruited, and eight were excluded, resulting in a final sample of 392. Participant age ranged from 18 to 80 years, with 53.5% male and 46.0% female. Most participants identified as White (78.5%), followed by Black or African American (9.0%) and Asian (8.2%). Household incomes were varied: 23.8% earned \$35,000–\$54,999, 15.9% earned \$15,000–\$34,999 or \$55,000–\$74,999, and 14.1% earned \$100,000–\$149,999. More than half of the participants (53.7%) held a college degree.

Table 1. Summary of mediation (Study 2) and moderated mediation (Study 3) analyses: Indirect effects, indices of moderated mediation, and direct effects on advertising outcomes.

Study 2	DV	Indirect effect (b)	BootSE	BootLLCI	BootULCI	Direct effect (b)	SE (HC4)	LLCI	ULCI
	$Att_{website}$	0.10	0.05	0.0186	0.2063	0.14	0.16	−0.1807	0.4692
	Att_{brand}	0.11	0.05	0.0186	0.224	0.17	0.19	−0.2086	0.5554
	PI	0.13	0.06	0.0230	0.2643	−0.12	0.19	−0.4965	0.2673

Study 3	DV	Index of moderated mediation (b)	BootSE	BootLLCI	BootULCI	Direct effect (b)	SE (HC4)	LLCI	ULCI
	$Att_{website}$	−0.06	0.03	−0.1164	−0.0125	0.12	0.10	−0.0816	0.3197
	Att_{ad}	−0.05	0.02	−0.0963	−0.0095	0.16	0.13	−0.1016	0.4134
	Att_{brand}	−0.05	0.02	−0.0979	−0.0096	0.09	0.13	−0.1645	0.3420
	PI	−0.06	0.03	−0.1298	−0.0133	0.19	0.15	−0.1101	0.4851

Note. Study 2 tested serial mediation using PROCESS Model 6; Study 3 tested moderated mediation using PROCESS Model 83. All analyses used 5,000 bootstrap resamples and HC4 standard errors.
Att_{ad} = attitude toward the ad; Att_{brand} = attitude toward the brand; $Att_{website}$ = attitude toward the ad; PI = purchase intention.

Experimental Design, Stimuli, and Procedure

Study 3 experimentally tested the effects of limited consent options—interfaces providing only an "I Agree" button without a clear opt-out alternative—on consumers' privacy perceptions and responses. This design operationalizes the most salient dark pattern identified in Studies 1 and 2, where participants consistently viewed the absence of a decline option as coercive and autonomy-reducing (Luguri and Strahilevitz 2021; Mathur, Kshirsagar, and Mayer 2021).

Participants completed a Qualtrics survey after providing electronic consent. The same fictional travel agency website used in Study 2 served as the experimental stimulus. Participants were randomly assigned to one of four conditions in a 2×2 factorial design: *consent option* (*I Agree Only* vs. *I Agree and Decline*) $\times$ *ad type* (*targeted* vs. *non-targeted*).

Following exposure to the website and data consent disclosure, participants viewed either a targeted or non-targeted ad. The targeted ad manipulation also allowed us to assess retro-cued accessibility, examining whether ad exposure prompted recall of the earlier consent experience. Measures included perceived privacy threat, perceived control, and attitudes toward the website, ad, and brand, as well as PI. Finally, participants completed demographic questions and were debriefed.

Measures

The measures from Study 2 were replicated, assessing attitudes toward the website ($M = 4.02$, SD = 1.38, ω = .93); ad ($M = 3.99$, SD = 1.47, ω = .93); brand ($M = 4.09$, SD = 1.45, ω = .94); and PI ($M = 3.17$, SD = 1.73, ω = .97) on a 7-point scale. To measure perceived threats to data privacy, we used four items ($M = 4.29$, SD = 1.56, ω = .92) on a 7-point Likert type scale. Perceived privacy control ($M = 3.71$, SD = 1.49, ω = .92) was measured using a four-item 7-point Likert type scale. Retrospective recognition of consent disclosures was assessed through three self-report items on a 7-point scale, asking: "How much did the ad help you remember the data consent notice on the travel agency website?" ($M = 4.38$, SD = 1.58, ω = .85). The items were: "*It helped me recall the cookie consent notice*"; "*It reminded me of the cookie consent notification*"; and "*I used it as a cue for the cookie consent message.*"[1]

Results

Manipulation and Randomization Checks

In the first pretest, we tested the ad targeting manipulation. Participants were randomly shown one of two social media-style ads: one featured a targeted ad promoting the travel agency website they had just visited, and the other displayed a non-targeted ad for an unrelated brand. Both ad formats were identical in visual layout and content, including a visible "Advertisement" label, brand name, and website URL (Appendix C). After viewing the ad, participants were asked to provide open-ended reflections about what they noticed or thought about the ad. These cognitive responses were qualitatively coded by two independent coders. Responses that referenced previous website visits or implied that the ad was shown based on browsing behavior were coded as indicating recognition of targeting (1); other responses were coded as not indicating such recognition (0). Inter-coder reliability was high (Perreault and Leigh's Ir = .83), and discrepancies were resolved through discussion. A chi-square test confirmed the effectiveness of the manipulation ($X^2(1, N = 80) = 49.57$, $p < .001$), showing that participants exposed to targeted ads were more likely to spontaneously recognize the relevance of the ad to their prior web activity.

In the second pretest, we assessed the manipulation of data consent options, comparing two interface conditions: one that included only an "I Agree" button and another that offered both "I Agree" and "Decline" options (Appendix B). Participants rated the extent to which they perceived the interface as using dark patterns on a 7-point scale. As expected, participants rated the "I Agree Only" condition as significantly more manipulative ($M = 4.82$, SD = 1.56) than the "I Agree and Decline" condition ($M = 4.10$, SD = 1.54; $F(1, 193) = 10.07$, $p = .002$, $\eta_p^2 = .05$).

In the main study, we employed equivalent methods for manipulating travel websites, except for the portion concerning data consent options (Appendix B). A satisfactory outcome was obtained from the randomization check for age ($F(1, 387) = 3.07$, $p > .05$); gender ($X^2(1, N = 391) = .86$, $p > .05$); education ($F(1, 387) = 3.52$, $p > .05$); income ($F(1, 387) = .29$, $p > .05$); and race ($X^2(1, N = 391) = .86$, $p > .05$), indicating the absence of systematic differences in these variables across the consent options and ad types.

Hypothesis Tests

We tested Hypothesis 2, which proposed that retrospective recognition of consent disclosure → perceived privacy threat → perceived control serially mediates the effect of consent option type on consumer responses. Descriptive means were similar across conditions (Hypothesis 2_a: $Att_{website}$ $M_{agree\text{-}only} =$

4.00 vs. $M_{agree+decline}$ = 4.05; Hypothesis 2_b: Att_{ad} = 4.02 vs. 3.96; Hypothesis 2_c: Att_{brand} = 4.08 vs. 4.09; Hypothesis 2_d: PI = 3.21 vs. 3.12). Although the direction of means was not uniform, such variation likely reflects boundary conditions and contextual nuances, as the theorized effects operate primarily through indirect pathways rather than direct mean differences.

We further examined whether exposure to a targeted ad (vs. non-targeted ad) moderated this mediation by enhancing cognitive accessibility—that is, by increasing retrospective recognition of the prior consent disclosure. Using PROCESS Model 83 with *heteroskedasticity-consistent* (HC4) standard errors, results revealed a significant interaction between consent option and ad type ($b = 0.83$, $SE = 0.33$, $t = 2.54$, p = .01, 95% CI [0.1868, 1.4675], ΔR^2 = .02, $F(1, 383)$ = 6.45, p = .01). When the ad was non-targeted, the effect of consent option on recognition was nonsignificant (b = −0.28, $SE = 0.22$, t = −1.26, p = .21, 95% CI [−0.7137, 0.1556]), but under targeted ad exposure, the effect was positive and significant ($b = 0.55$, $SE = 0.24$, $t = 2.31$, p = .02, 95% CI [0.0815, 1.0147]). These results indicate that targeted ads strengthened the link between consent option structure and recognition, consistent with the memory cueing and accessibility processes predicted by the proposed framework (see Figure 2).

For the moderated mediation analyses, all indirect effects were significant and consistent with theoretical predictions, while direct effects were non-significant—indicating indirect-only mediation. The indices of moderated mediation for each dependent variable confirmed the hypothesized pattern of effects. Analyses employed 5,000 bootstrap resamples with 95% CIs using PROCESS Model 83 and HC4 *heteroscedasticity-consistent* adjustments (Cribari-Neto).

The findings indicated that the impact of data consent options (1 = Agree only, 0 = Agree and Decline) on $Att_{website}$ was mediated consecutively through retrospective recognition, perceived threats to data privacy, and perceived data privacy control, with the indirect effect being particularly notable when encountering a targeted ad (b = −0.06, BootSE = .03, 95% CI [−0.1116, −0.0107]), excluding zero. The direct effect of consent options on $Att_{website}$ was not significant (b = .11, SE(HC4) = .10, 95% CI [−0.0830, .3121]), including zero. The results showed *indirect-only mediation*. To be specific, when a non-targeted ad was shown, the consent option effect was not significant (b = −0.28, SE(HC4) = .22, t = −1.26, p > .05, 95% CI [−0.7074, .1546]), whereas when a targeted ad was shown, the consent option without a

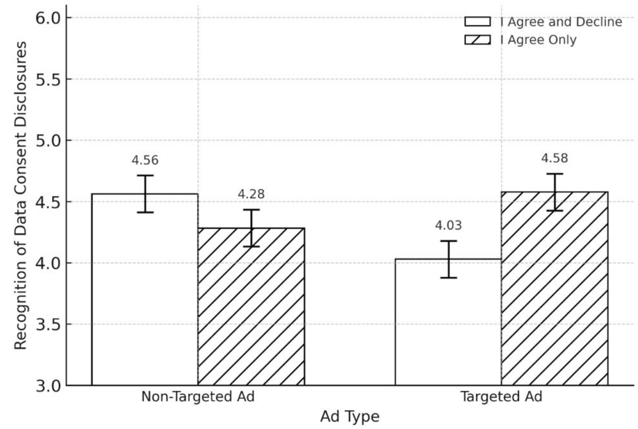

Figure 2. The moderating effect of exposure to a targeted ad. *Note.* Mean recognition of data-consent disclosures by ad type (targeted vs. non-targeted) and consent option (I Agree Only vs. I Agree and Decline). Error bars represent ± 1 standard error for descriptive illustration. Inferential results are reported in the text based on bootstrapped moderated-mediation analyses (PROCESS Model 83).

decline choice enhanced retrospective recognition (b = .52, SE(HC4) = .23, $t = 2.23$, p = .02, 95% CI [.0602, .9705]). As a result, the indirect effect (*retrospective recognition of disclosure-data privacy threats-perceived control*) was significant in the targeted ad condition (b = −0.04, BootSE = .02, 95% CI [−0.0813, −0.0062]), but not in the non-targeted ad condition (b = .02, BootSE = .02, 95% CI [−0.0128, .0546]).

Similarly, for attitudes toward the ad, the moderated mediation index was significant (b = −.05, BootSE = .0222, 95% CI [−.0963, −.0095]). The direct effect of consent options on ad attitudes was not significant (b = .16, SE(HC4) = .13, p = .23, 95% CI [−.1016, .4134]). For attitudes toward the brand, the moderated mediation index was also significant (b = −.05, BootSE = .0226, 95% CI [−.0979, −.0096]), with a non-significant direct effect (b = .09, SE(HC4) = .13, p = .49, 95% CI [−.1645, .3420]). Finally, for PI, the index of moderated mediation was significant (b = −.06, BootSE = .03, 95% CI [−.1298, −.0133]), and the direct effect was not significant (b = .19, SE(HC4) = .15, p = .22, 95% CI [−.1101, .4851]).

The results showed that viewing a targeted ad increased recognition of the consent disclosure (Figure 2). When paired with the absence of a refusal option, this recognition heightened perceived data privacy threats and reduced perceived control. Overall, limited consent architectures—representing dark pattern designs—amplified privacy concerns and lowered perceived autonomy, leading to more negative evaluations of the website, ad, brand, and PI. These results provide strong support for Hypothesis 2_{a-d}. The moderated mediation and direct effects are summarized in Table 1.

Study 3 Discussion

Study 3 conceptually replicated Study 2 and extended its findings by showing that persuasion knowledge, heightened through exposure to targeted ads, activates psychological reactance and shapes ad responses. The results illustrate how consumers adjust their privacy management strategies in response to both internal cognitive processes and external contextual cues. Whereas non-targeted ads did not affect recall of prior consent experiences, targeted ads enhanced recognition of consent disclosures lacking a decline option, thereby activating persuasion knowledge. This recognition, coupled with restricted privacy choices, heightened awareness of manipulative design and intensified reactance. Overall, the findings advance a dynamic account of consumer privacy protection, showing that resistance to manipulative consent practices depends not only on recognizing persuasion tactics but also on maintaining perceived control and recalling prior consent experiences.

General Discussion

This research integrates PKM, PRT, and cognitive accessibility to explain how dark patterns in OBA consent disclosures shape consumer evaluation. Ambiguous wording and limited choice heighten perceived privacy threats and reduce control, while targeted ads act as retrieval cues reactivating prior consent experiences. These effects clarify how consent design and advertising cues jointly trigger persuasion knowledge and reactance, advancing a dynamic account of consumer coping in data-driven persuasion contexts.

Theoretical Implications

Despite the ubiquity of OBA, the impact of its consent disclosures on consumer coping strategies remains underexplored. Examining consumer data privacy rights within increasingly complex digital dark patterns (FTC 2022) is both timely and essential. Earlier research often portrayed consumers as passive recipients of persuasion who react defensively when manipulation is detected (Isaac and Grayson 2017). More recent perspectives describe consumers as active agents who evaluate persuasion attempts through contextual and transparency cues (Boerman, Van Reijmersdal, and Rozendaal 2024; Ham 2017; Shin and Yu 2021). Activation of persuasion knowledge can lead to resistance or acceptance depending on whether persuasive efforts are perceived as ethical and fair (Evans, Wojdynski, and Grubbs Hoy 2019).

Integrating PKM and PRT, this research demonstrates that dark patterns in OBA consent disclosures, particularly ambiguous language and restricted choice architecture, heighten perceived threats to data privacy and reduce perceived control, which produces psychological reactance. In contrast, transparent consent designs that use clear language and provide explicit decline options enhance perceptions of fairness and autonomy, strengthening trust and reducing reactance. Targeted advertising, conceptualized as a contextual factor, shapes how consumers interpret these consent experiences. When consumers encounter targeted ads after manipulative consent interfaces, targeting cues reactivate prior consent memories and make manipulative intent more cognitively accessible, intensifying persuasion knowledge and reactance (Acquisti, Brandimarte, and Loewenstein 2015; Kim, Barasz, and John 2019; White et al. 2008). However, when targeted ads follow transparent consent experiences, consumers interpret the targeting context as consistent with their expectations, reducing defensive motivation.

The unexpected findings offer further theoretical insight into this dynamic. When participants viewed non-targeted ads after transparent consent interfaces that included both "I Agree" and "Decline" options, they showed unexpectedly high recognition of the consent disclosure. This finding indicates that transparency enhanced memory encoding through perceived fairness and autonomy, prompting greater attentional engagement and metacognitive processing (Campbell and Kirmani 2000). In contrast, when targeted ads followed transparent consent, disclosure recognition was lowest. This outcome is not simply a matter of perceived norm compliance but reflects cognitive congruence between the transparent consent and the relevant ad. When both stimuli aligned conceptually, the absence of dissonance or novelty reduced elaboration and memory retrieval. Conversely, when a non-targeted ad followed a transparent consent experience, the incongruent stimulus introduced a cognitive mismatch that reactivated disclosure memory and strengthened recognition. This pattern aligns with associative network and contrast-based elaboration mechanisms in memory theory

(Anderson 1983; Keller 1987) and illustrates how contextual incongruity can heighten accessibility through deeper processing.

While recognition of the disclosure is itself a function of accessibility, the present research extends theory by showing that accessibility alone does not explain subsequent persuasion outcomes. Once the disclosure is cognitively reactivated, consumers engage in evaluative judgment about the persuasive technique, consistent with the meta-cognitive dimension of PKM. The integrative framework developed here demonstrates that cognitive accessibility (from associative memory theory) and evaluative appraisal (from PKM) operate sequentially. Accessibility allows recognition of prior persuasion episodes, whereas evaluative judgment determines whether consumers resist or accept the subsequent message. This process highlights how persuasion knowledge is not only stored and retrieved but also interpreted and applied in real time—a dynamic that neither associative network models nor PKM alone fully capture.

This study therefore advances PKM, PRT, and cognitive memory research (Anderson 1983; Souza and Oberauer 2016) by conceptualizing persuasion knowledge accessibility as a precursor to evaluative activation. The associative network model explains how contextual overlap or contrast between targeted ads and prior consent experiences facilitates retrieval, consistent with retro-cue effects in cognitive psychology (Souza and Oberauer 2016). Our findings align with evidence that retrieval cues consistent or inconsistent with stored experiences can enhance recognition depending on salience (Boerman, Kruikemeier, and Zuiderveen Borgesius 2017; Grigsby and Mellema 2020). They also support arguments that facilitating cognitive access to persuasion knowledge empowers consumers to identify and evaluate manipulative intent more effectively (Eisend and Tarrahi 2022; Packard, Gershoff, and Wooten 2016).

Extending recent PKM work (Boerman, Kruikemeier, and Zuiderveen Borgesius 2021, Boerman, Van Reijmersdal, and Rozendaal 2024; Ham and Kim 2019), this study identifies accessibility as the cognitive route through which consent design and advertising context jointly shape persuasion outcomes. Although individual differences were not directly tested, prior research suggests these processes may vary based on cognitive capacity, motivation (Campbell 1995), privacy literacy, and regulatory awareness. Future studies should examine how these factors, along with privacy concern and reactance proneness, influence the encoding–retrieval–evaluation sequence in data-driven persuasion contexts.

Overall, consumer coping in OBA reflects two interconnected cognitive routes. Encoding valence, influenced by fairness and clarity in consent design, determines the strength of memory traces. Retrieval cueing, prompted by targeting cues and contextual contrast, determines whether those memories are reactivated and further evaluated. The recognition of persuasive techniques and subsequent evaluation complete the process by determining consumer acceptance or resistance. By explaining how dark patterns interact with targeting cues through accessibility-based and evaluative mechanisms, this research provides a unified theoretical framework that integrates privacy, persuasion, and cognition, offering new insight into ethical design and consumer protection in digital marketing.

Managerial and Public Policy Implications

Data protection regulations are crucial for safeguarding consumer data privacy and building trust with consumers. However, dark patterns in marketing tactics and psychological strategies have long been employed (FTC, 2022). The recent FTC (2022) report highlights the proliferation of manipulative design practices, making it difficult for consumers to deal with such inappropriate tactics. Both FTC advertising regulations and data privacy laws like CCPA are increasingly focusing on addressing these challenges to safeguard consumer privacy rights in today's evolving data landscape. Without clear guidance for OBA advertisers and digital publishers, targeted OBA raises social concerns. Digital advertisers must prioritize consumer privacy, investing in resources and technologies to uphold data rights. Enhancing transparency and clarity in data consent disclosures is an essential step toward this goal. The FTC (2022) stressed that consent must be clearly defined to ban deceptive tactics because some marketers may provide disclosures without securing informed, voluntary, clear, and specific agreement. Consumers are often bombarded with confusing consent practices that fail to offer meaningful choices. Therefore, the recent update to the CCPA—the California Privacy Rights Act (CPRA)—explicitly bans the use of dark patterns in obtaining consumer consent (California Privacy Rights Act 2020).

Our findings align with recent research on privacy concerns. Acquisti, Brandimarte, and Loewenstein (2015) argued that both commercial and governmental entities have developed expertise in exploiting psychological and behavioral processes to promote personal information disclosure, often through manipulative practices such as "malicious interface design" (512). The concept of *malicious interface design* refers to design features—such as dark patterns in data consent forms—that deliberately confuse users into disclosing personal information. Because such deceptive designs are difficult for consumers to detect, individuals face significant challenges in protecting themselves from involuntary information sharing. As Acquisti, Brandimarte, and Loewenstein (2015) emphasized, merely providing a consent form is insufficient.

Upon this perspective, Brandimarte, Acquisti, and Loewenstein (2013) introduced and empirically demonstrated the *privacy paradox*, which occurs when individuals believe they have sufficient control over their privacy—although this perception is inaccurate—leaving them more vulnerable because they are more likely to disclose personal information under the false sense of protection. Although our study does not directly test the privacy paradox, we concur with their argument that consent mechanisms alone not only fail to adequately protect consumers' privacy but may also foster a misleading perception of security. Given that consumers' decisions are heavily influenced by contextual factors and interface designs, including dark patterns (Acquisti, Brandimarte, and Loewenstein 2015), stronger baseline protections are essential to prevent shifting the full burden of privacy protection onto individuals. Building sustainable trust in the digital advertising ecosystem requires collective, society-wide engagement that extends beyond consumers to include regulators, publishers, platforms, and advertisers. Strict compliance with privacy laws and proactive adoption of industry standards are essential to fostering consumer confidence. Theory-driven and empirically supported data privacy regulations can ensure that user interfaces are designed to enhance transparency, accessibility, and user-friendliness, especially for those with limited data privacy literacy. By addressing data consent disclosure language and user interface design comprehensively, policymakers can foster a more secure and fair digital environment for all consumers. Future research should further examine how such collaborative approaches to transparency influence trust in consent processes, consumer attitudes, and ultimately the effectiveness—or potential risks—of targeted OBA.

While regulatory initiatives remain essential, enhancing data privacy literacy can provide an equally important complement. Because data practices continually evolve, it is unrealistic to expect foolproof regulation. Instead, individuals must adapt to changing contexts and develop the skills to manage emerging data-use cases effectively (Boerman, Kruikemeier, and Zuiderveen Borgesius 2017; Ham 2017). In this regard, consumers can leverage their knowledge to respond critically to firms that neglect privacy needs. Research shows that advertising literacy—such as recognizing selling intent—helps even children cope with persuasive messages (Nelson 2016). Similarly, cultivating data privacy literacy through formal education and organizational training is vital. Such efforts not only empower consumers to make informed choices but also highlight the importance of interface design in activating persuasion knowledge within digital media environments.

Limitations and Suggestions

Several limitations should be acknowledged, each of which provides meaningful directions for future inquiry. First, to strengthen external validity, future research should examine a broader range of website contexts beyond the controlled experimental setting employed here. Replicating these findings across diverse digital platforms and consent environments would help assess the generalizability of the observed psychological mechanisms.

Second, this study did not explicitly account for individual dispositional differences that may shape how consumers process data consent and targeted advertising cues. Factors such as cognitive motivation and capacity are likely to moderate the accessibility and activation of persuasion knowledge (Boerman, Kruikemeier, and Zuiderveen Borgesius 2021; Boerman, Van Reijmersdal, and Rozendaal 2024). Future work should investigate how internal (e.g., dispositional) and external (e.g., message-driven or situational) accessibility cues interact to influence persuasion knowledge activation, offering a more comprehensive understanding of the boundary conditions of these effects.

Third, although our studies revealed statistically significant indirect effects through theoretically grounded serial mediation pathways, mean differences across experimental conditions on some distal outcome variables were less consistent. These patterns do not undermine the theoretical model but rather underscore that the psychological mechanisms—specifically, perceived threat and perceived control—operate subtly and cumulatively. Theoretically, this aligns with contemporary perspectives emphasizing that significant indirect effects can meaningfully demonstrate mediation even when total effects are nonsignificant (Hayes 2009, 2015, 2018; Preacher and Hayes 2004;

Zhao, Lynch, and Chen 2010). As mediation reflects underlying process rather than magnitude, our findings provide evidence that dark pattern tactics influence consumer responses through specific psychological pathways rather than broad evaluative shifts. Nonetheless, future studies should continue to test these mechanisms under varied conditions and populations to establish robustness and ecological validity.

Beyond these methodological considerations, broader implications arise at the regulatory and societal levels. Around the world, and increasingly within the United States, data privacy regulations such as the CCPA and GDPR continue to evolve. As the digital economy transcends national boundaries, marketers in the United States must ensure compliance with GDPR, whereas European firms must increasingly navigate U.S. state-level privacy laws. Given the rapid evolution of these frameworks, scholarly research often struggles to keep pace with emerging regulatory and technological realities. Greater collaboration between policymakers, scholars, and industry stakeholders is essential to close this gap and ensure evidence-based data governance.

Finally, while this study applied the PKM framework within the context of OBA consent disclosures, future research should extend its application to broader data privacy and consumer–platform interactions. Today's digital journey is fluid and multitouchpoint, involving numerous actors beyond a single publisher or advertising platform. Thus, future research should adopt an ecological perspective that captures how persuasion knowledge and privacy perceptions unfold dynamically across the consumer's full digital experience. Such an integrated approach, though methodologically demanding, can offer a more holistic understanding of how consent design, interface cues, and advertising practices jointly shape consumer trust and autonomy.

Conclusion

This research advances advertising theory and practice by providing an integrated and empirically grounded account of how consumers recognize and respond to dark patterns in data consent disclosures within OBA. Across three studies, we show that linguistic framing and choice architecture influence perceptions of privacy threat and control, activating cognitive and evaluative mechanisms that shape persuasion outcomes. Drawing on PKM, PRT, and cognitive accessibility, this work clarifies how consent design and advertising context jointly affect consumer judgment and resistance. These insights emphasize that responsible digital persuasion depends less on technical sophistication than on transparent, autonomy-supportive design. As data privacy becomes central to the future of marketing communication, this study offers a theoretical foundation for developing advertising systems that respect consumers' cognitive agency and reinforce trust in the digital marketplace.

Note

1. An exploratory factor analysis using principal axis factoring was conducted for the retrospective recognition of data consent disclosure. The Kaiser-Meyer-Olkin value was .62 and Bartlett's test of sphericity was significant ($\chi^2(3) = 707.95$, $p < .001$), indicating factorability. One factor was extracted, explaining 68.58% of the variance. All items loaded strongly on a single factor (loadings = .54 to .98), supporting unidimensionality. A follow-up confirmatory factor analysis further validated this measurement model. The model showed acceptable fit ($\chi^2(1) = 2.81$, $p = .09$; CFI = .99; RMSEA = .07; SRMR = .03).

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Data Availability Statement

The data underlying this article will be shared on reasonable request to the corresponding author.

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